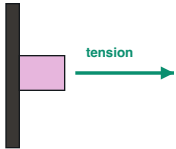


WHY NOT JUST MAGNETS?

"Amazon has magnets that can handle that weight fine, and are cheap — why can't we just use those?"

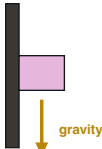
A fair question, and the answer is arithmetic, not opinion. It takes three steps: what a rating measures, what the fridge actually does to it, and what a lever does to whatever is left.

PULL — what the box quotes



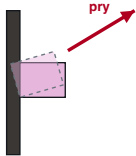
Straight away from the surface. Ratings are measured THIS way, on thick ground steel at zero gap. Strongest direction by far.

SHEAR — what gravity applies



Sliding DOWN the panel. The magnet only resists through friction: grip = pull x mu, and mu is tiny on smooth paint. Weakest direction.

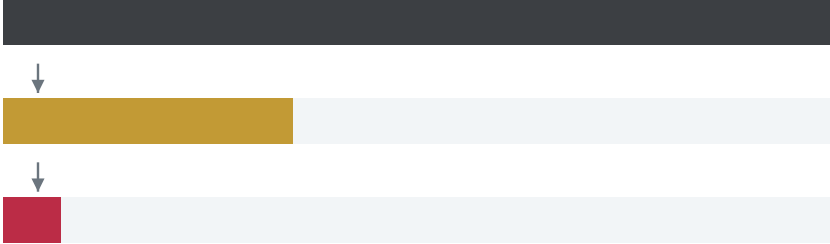
PEEL — what a bump applies



Levering one edge away. The prying force is multiplied by the lever, and it lands on ONE magnet at a time — never on the total.

WATCH THE NUMBER ON THE BOX COLLAPSE

The as-built magnet is a 48 mm (1.89 in) pot magnet, and its catalogue rating is honest — for the test it describes. The fridge is not that test.



RATED 175 lb (778 N)
measured on THICK ground steel, ZERO gap, pulling STRAIGHT OFF

x 0.35 on the fridge = 61.2 lb (272 N) PULL
0.6-0.9 mm painted sheet: thin steel saturates, paint holds the magnet off the metal

x mu 0.2 friction = 12.2 lb (54 N) SHEAR
the load is SLIDING, not pulling — bare nickel on smooth paint barely grips

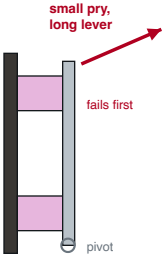
12.2 lb (54 N) is 7% of the number on the box — in the direction the fridge actually loads it.

Not our theory — the vendors publish the drop themselves.

totalElement rate their 43 mm rubber pot magnet at 33.5 lb (149 N) vertical pull but only 7.0 lb (31 N) horizontal — 21% of the headline, from their own catalogue.

PEEL: YOU NEVER FIGHT THE TOTAL

Four magnets at 61 lb (272 N) each sounds like 245 lb (1090 N) of holding. It is not. Bump the screen's top edge and the plate rotates about its bottom — the prying force is multiplied by the lever ratio, and the whole moment lands on the TOP magnet alone, at its own 61 lb. The instant it lets go, its share dumps onto the next one down. Magnet mounts do not fail by being overpowered; they fail by unzipping, one magnet at a time.



WHAT MAGNET-ONLY WOULD ACTUALLY NEED

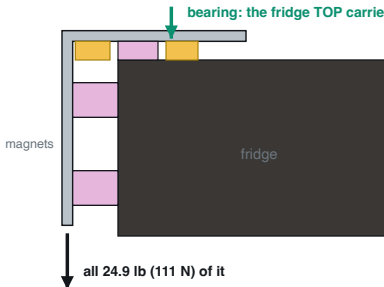
Hanging load: display 8.7 lb (39 N) + bracket 16.2 lb (72 N)
= 24.9 lb (111 N)

Rated pull needed per pound held, in shear: 1 / (0.35 x 0.2)
= 14 lb rated per lb held

So holding 24.9 lb (111 N) in shear needs
~238 lb (1.1 kN) RATED — 14x whatever it holds, before any safety factor

And if the panel is 304 stainless (many are), every magnet contributes
exactly ZERO. This one measured magnetic — the design still must not care.

THE ANSWER: DON'T FIGHT THE PHYSICS — HOOK OVER THE TOP



The arm rests ON the fridge top. The vertical load goes into bearing at the corner — compression on painted steel over a sponge pad. It never touches bare steel: the pad and the arm magnets, both about 11.5 mm, hold it clear.

The magnets carry ZERO vertical load. They only stop swing and rattle, and that duty is TENSION — their strong direction.

Demand on a magnet: 1.6 lb (7 N) from a firm 5 lb (22 N) touch press, 2.2 lb (10 N) of total peel — against 61 lb (272 N) of derated capacity each: 37x in hand.

A non-magnetic panel no longer matters: the hook still carries everything, and the arm's width still resists the twist.

The stakes: 3.94 kg (8.7 lb) of optically bonded glass over a kitchen floor. "Probably strong enough" is not a spec — a hook is.

SO HOW STRONG WOULD THEY HAVE TO BE? — the fair answer

A magnet-only mount needs no arm, so it is lighter: screen 3.94 kg + body plate 2.80 kg + 4 magnets = 16.6 lb (74 N), not 24.9 lb (111 N).
To hold THAT by friction alone, TOTAL rated pull across all magnets (divide by 0.070):

SF 1.0	on the edge of sliding	238 lb (1057 N)	= 30 lb x 8 magnets
SF 2.5	project standard	594 lb (2642 N)	= 74 lb x 8 magnets
SF 4.0	overhead glass	950 lb (4227 N)	= 119 lb x 8 magnets

The magnets already specified are 175 lb each. Eight of them give 98.0 lb (436 N) of shear — SF 5.9 on that load.

So shear is NOT the reason for the hook. Magnets can carry this weight. The hook is there for what shear numbers do not cover:

PEEL — a lever at one corner beats a single magnet, not the total, and the rest unzip after it.
A NON-MAGNETIC panel makes every one of those numbers zero. This one measured magnetic; the design still must not depend on it.
CREEP — the paint is a polymer and it flows under sustained shear, while the compressor vibrates the panel around the clock. Every cycle lets the magnet micro-slip, always downhill. An SF that holds today need not hold in a year.
The failure mode is bonded glass onto a kitchen floor, with no warning.